

Evaluating the Solow Growth Model

Balanced growth, convergence, and international income differences

Questions for evaluating the model

1. Does the model reproduce the balanced-growth properties observed in developed economies?
2. Does it predict convergence in income per worker?
3. Can plausible differences in saving, population growth, and technology explain international income gaps?

Output per person grows at the rate of technological progress

The production function in efficiency units is

$$y = A\hat{y} = A\hat{k}^\alpha, \quad \hat{y} = \hat{k}^\alpha.$$

Therefore,

$$\frac{\dot{y}}{y} = \frac{\dot{A}}{A} + \frac{\dot{\hat{y}}}{\hat{y}} = x + \alpha \frac{\dot{\hat{k}}}{\hat{k}}.$$

The law of motion for capital per effective worker implies

$$\frac{\dot{\hat{k}}}{\hat{k}} = s\hat{k}^{\alpha-1} - (\delta + n + x), \quad \frac{\partial(\dot{\hat{k}}/\hat{k})}{\partial \hat{k}} = (\alpha - 1)s\hat{k}^{\alpha-2} < 0.$$

Long-run implication

As $\hat{k}$ approaches its steady state, $\dot{\hat{k}}/\hat{k} \rightarrow 0$. Hence $\dot{y}/y \rightarrow x$.

Consumption and investment display balanced growth

The resource constraint and constant saving rate imply

$$C = (1 - s)Y, \quad I = sY.$$

Because s remains constant,

$$\frac{\dot{C}}{C} = \frac{\dot{Y}}{Y}, \quad \frac{\dot{I}}{I} = \frac{\dot{Y}}{Y}.$$

Prediction

In the long run, output, consumption, and investment per person all grow at the rate x .

This property broadly describes developed economies. In many developing economies, however, saving rates rise during development.

Capital grows faster than labour in the long run

Since capital per effective worker equals

$$\hat{k} = \frac{K}{AL},$$

its growth rate satisfies

$$\frac{\dot{K}}{K} = \frac{\dot{A}}{A} + \frac{\dot{L}}{L} + \frac{\dot{\hat{k}}}{\hat{k}} = x + n + \frac{\dot{\hat{k}}}{\hat{k}}.$$

Along the balanced growth path, $\dot{\hat{k}}/\hat{k} = 0$, so

$$\boxed{\frac{\dot{K}}{K} = n + x}$$

and

$$\boxed{\frac{\dot{K}}{K} - \frac{\dot{L}}{L} = x}.$$

The rate of return to capital is constant

With competitive factor markets, the net return to capital is

$$r - \delta = \alpha(AL)^{1-\alpha}K^{\alpha-1} - \delta = \alpha\hat{K}^{\alpha-1} - \delta.$$

Along the balanced growth path, $\hat{k}$ is constant. Therefore,

$r - \delta \text{ is constant in the long run.}$

Although K and AL both grow, their effects on the marginal product of capital in the long run offset each other.

Long-run growth and transitional growth

From $y = A\hat{k}^\alpha$,

$$\frac{\dot{y}}{y} = \underbrace{x}_{\text{long-run growth}} + \underbrace{\alpha \frac{\dot{\hat{k}}}{\hat{k}}}_{\text{transitional growth}}.$$

The transition term obeys

$$\frac{\dot{\hat{k}}}{\hat{k}} = s\hat{k}^{\alpha-1} - \delta - n - x, \quad \frac{\partial(\dot{\hat{k}}/\hat{k})}{\partial \hat{k}} < 0.$$

Key mechanism

An economy with low capital per effective worker has a high marginal product of capital and grows rapidly during its transition to the steady state.

Conditional convergence

Suppose a country is poor because its initial K/L is low, while s , n , x , δ , and the production technology A are the same across countries.

$$\hat{k} \text{ low} \implies Y/L \text{ low} \quad \text{and} \quad \dot{\hat{k}}/\hat{k} \text{ high} \implies \dot{y}/y \text{ high.}$$

Conditional convergence

Countries converge to the same steady state when the parameters that determine that steady state are the same.

Why absolute convergence need not occur

If a country is poor because its technology level A is low, then its capital per effective worker may be high:

$$\hat{k} = \frac{K}{AL}.$$

A lower A can therefore imply

$$Y/L \text{ low}, \quad \hat{k} \text{ high}, \quad \dot{\hat{k}}/\hat{k} \text{ low}, \quad \dot{y}/y \text{ low}.$$

Differences in s , n , or x also produce different steady states. A poor country need not grow faster than a rich country.

Prediction

The Solow model predicts conditional convergence, but not convergence among all countries regardless of their structural characteristics.

The model can also predict divergence

Consider two countries that initially have the same income per worker.

- ▶ If one country has a higher saving rate, it converges to a higher steady-state income level.
- ▶ If one country has faster technological progress, its long-run growth rate is higher.

Their income paths diverge even though they begin at the same income level.

Empirical interpretation

The model allows convergence within groups of similar countries or regions, while permitting persistent or widening gaps across dissimilar economies.

Transitional growth - a numerical example

Suppose

$$s = 0.25, \quad \alpha = \frac{1}{3}, \quad \delta = 0.05, \quad n = 0.01, \quad x = 0.02.$$

If $\hat{k}_0 = 1.8$, the model predicts initial growth of output per worker near 5%:

$$\frac{\dot{y}}{y} = x + \alpha \left[s \hat{k}^{\alpha-1} - (\delta + n + x) \right] \approx 0.05.$$

Implication

Rapid growth can arise when an economy begins far below its steady state, but the transitional component fades as capital accumulates.

Transitional growth falls toward its long-run rate

t	$\hat{k}_t$	$\dot{\hat{k}}_t/\hat{k}_t$	$\dot{y}_t/y_t$	t	$\hat{k}_t$	$\dot{\hat{k}}_t/\hat{k}_t$	$\dot{y}_t/y_t$
0	1.8000	—	—	7	2.8267	0.0491	0.0364
1	1.9601	0.0890	0.0497	8	2.9541	0.0451	0.0350
2	2.1162	0.0796	0.0465	9	3.0764	0.0414	0.0338
3	2.2678	0.0717	0.0439	10	3.1939	0.0382	0.0327
4	2.4149	0.0648	0.0416	11	3.3066	0.0353	0.0318
5	2.5571	0.0589	0.0396	12	3.4145	0.0326	0.0309
6	2.6944	0.0537	0.0379	13	3.5178	0.0303	0.0301
				14	3.6166	0.0281	0.0294

Output growth falls quickly toward the long-run rate of 2%.

Limits of transitional growth

The model has difficulty sustaining a growth miracle for several decades:

$$\frac{\dot{y}}{y} = x + \alpha \frac{\dot{\hat{k}}}{\hat{k}}, \quad \frac{\dot{\hat{k}}}{\hat{k}} \longrightarrow 0.$$

- ▶ Sustained growth near 5% eventually requires technological progress near 5%.
- ▶ A prolonged growth disaster similarly requires technological progress near zero or below zero.

Capital accumulation can generate large short-run growth differences, but technology determines long-run growth.

Steady-state income per worker

In the steady state,

$$0 = s\hat{k}^\alpha - (\delta + n + x)\hat{k}.$$

Solving for capital per effective worker gives

$$\hat{k}^* = \left(\frac{s}{\delta + n + x} \right)^{1/(1-\alpha)}.$$

Since $Y/L = A\hat{k}^\alpha$,

$$\left(\frac{Y}{L} \right)^* = A \left(\frac{s}{\delta + n + x} \right)^{\alpha/(1-\alpha)}.$$

Long-run income depends on technology and on the parameters that determine steady-state capital intensity.

Can saving rates explain a tenfold income gap?

Suppose rich and poor countries share A , n , x , δ , and α , but differ in saving rates. Then

$$\frac{y_R}{y_P} = \left(\frac{s_R}{s_P} \right)^{\alpha/(1-\alpha)}.$$

With $y_R/y_P = 10$ and $\alpha = 1/3$,

$$\frac{s_R}{s_P} = 10^{(1-\alpha)/\alpha} = 10^2 = 100.$$

If $s_R = 0.25$, this requires

$$s_P = \frac{0.25}{100} = 0.0025.$$

Assessment

Poor countries tend to save less, but a saving rate of 0.25% is implausibly low. Saving differences explain only part of the income gap.

Can population growth explain the income gap?

Holding A and s constant,

$$\frac{y_R}{y_P} = \left(\frac{\delta + n_P + x}{\delta + n_R + x} \right)^{\alpha/(1-\alpha)}.$$

For $y_R/y_P = 10$ and $\alpha = 1/3$,

$$n_P = 100(\delta + n_R + x) - \delta - x.$$

Using $\delta = 0.05$, $n_R = 0.01$, and $x = 0.02$ gives

$$n_P = 7.93,$$

or approximately 793% per year.

Assessment

Population-growth differences of plausible size cannot explain a tenfold income gap.

Can technology explain the income gap?

If the remaining parameters are the same across countries, then

$$\frac{y_R}{y_P} = \frac{A_R}{A_P}.$$

A tenfold income gap therefore requires

$$\frac{A_R}{A_P} = 10.$$

Differences in technology or total factor productivity can, in accounting terms, generate large differences in income per worker.

Limitation

The basic Solow model treats the technology level as exogenous. It does not explain why technology differs across countries.

Overall assessment of the Solow model

- ▶ It reproduces key balanced-growth facts for developed economies.
- ▶ It predicts conditional convergence through diminishing returns to capital.
- ▶ Capital accumulation can generate temporarily high growth, but not permanently high growth.
- ▶ Plausible saving and population-growth differences explain only part of international income gaps.
- ▶ Technology can account for large income differences, but the model leaves technology exogenous.

Central lesson

The Solow model organizes the proximate sources of growth and income differences, while leaving their deeper causes outside the model.